

Pranoy Radhakishnan

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EXPERIENCE

Avataar AI, Bangalore

March 2026 - Present

AI Engineer

- **Audio Compliance POC:** Developed an audio analysis proof of concept using **Whisper** for speech transcription, **PANNs** for acoustic event detection, **Gemma** for FTC compliance detection, and **Demucs** for signal-to-noise ratio (SNR) detection and audio separation.

Aqxle AI, Bangalore

Feb 2025 - Feb 2026

Senior AI Research Engineer

- Building AI Agents for the marketing flywheel that helps marketers & marketing/media researchers
- Developed custom agents for automated survey data processing, integrating natural language chat interfaces for dynamic insight generation and graph-based visualization.
- Explored and implemented automated EDA methods for faster and more consistent data analysis.

Accessia, Bangalore

April 2024 - Jan 2025

Computer Vision Engineer

- Led computer vision development focused on ML pipeline optimization in C/C++ and CUDA
- Implementing structured pruning, post-quantization, and quantization-aware training (QAT) for high-performance model compression.

First Sense, Bangalore (later acqui-hired by Accessia)

Nov 2021 - Feb 2024

Computer Vision Engineer

- Led the Computer Vision team for *CUBO*, developing animatable 3D morphable models (3DMMs) for MOOCs and online learning platforms using pose and shape estimation techniques.
- Engineered visual enhancement solutions, including whiteboard clarity improvement for virtual classrooms.
- Implemented advanced vision models, such as a modified Detection Transformer (DETR) for 2D human pose estimation, optimizing accuracy and performance.

Neuroplex, Trivandrum

Aug 2018 - Jan 2021

Co Founder/Computer Vision Engineer

- Led the team and developed ML models for *NAZAR*, a Video Content Analysis platform deployed across Indian Oil petrol pumps to monitor customer interactions, analyze customer flow, and enhance safety and security using existing CCTV infrastructure.
- Fine-tuned and deployed computer vision models on edge devices (NVIDIA Jetson Nano, NX Xavier) for real-time inference and scalability.
- Implemented and optimized algorithms for human and vehicle tracking, license plate recognition, vehicle re-identification, image segmentation, human action recognition, and attribute detection.

OPEN SOURCE PROJECTS

Active contributor to lucidrains repositories, implementing research papers and improving codebases across diffusion models, flow matching, video-action learning, VLA systems, and hierarchical RL.

- [Vit-Pytorch](#) : Implemented ViT-5 and JetViT
- [Transfusion-PyTorch](#) : Implemented unconditional KV caching, matching null-token sequence length to prompts for correct CFG alignment, and adding classifier-free guidance.
- [Rectified Flow-PyTorch](#) : Implemented UA Flow and LapLow
- [Mimic-Video](#) : Improved noise scheduling, VAE input normalization, and action decoding with proprioceptive masking.
- [MetaController](#) : Fixed Switching Unit logic by incorporating the previous subgoal (z_{t-1}) for correct hierarchical control.
- [Pi-Zero-PyTorch](#) : Fixed CFG RoPE alignment, temporal buffering, device compatibility, and unconditional pass robustness.

Past Personal Projects: Implemented, trained, and benchmarked deep learning models across video classification, object detection, pose estimation, and transformer-based vision systems.

- [Cnn-lstm](#) : CNN-LSTM architecture for video classification
- [Pix2seq](#) : Autoregressive transformer approach that formulates vision tasks as sequence prediction
- [Attention-models](#) : Implementations of various transformer architectures
- [Multi-instance-pose-estimation](#) : Multi-person human pose estimation using Transformers

TECHNICAL SKILLS

- **Languages & Frameworks** : Python, PyTorch, TensorFlow, Keras, C/C++, CUDA, LangChain
- **ML Systems** : Diffusion / Flow Matching, Transformers, Agentic Architectures
- **Optimization & Deployment** : ONNX, TensorRT
- **Databases** : Neo4j, MongoDB, MySQL
- **Mathematics** : Linear Algebra, Multivariable Calculus, Probability & Statistics.

EDUCATION

Indian Institute of Technology, Madras (Part Time) 2022 - Current

Bachelor of Science in Data Science and Programming

Kerala University, Trivandrum

Coursework in Electronics & Communication Engineering 2012 - 2016

AWARDS

- **Best Project of the Year** — Rule-based conversational robot
- **2nd Place**, Cyberdome Kerala Competition (Kerala Police)
- **IOCL Startup Funding** — Nationwide selection for AI video analytics platform